

Certificate of participation

Issued to : Filip-Alexandru Mocanca

Title of the course :

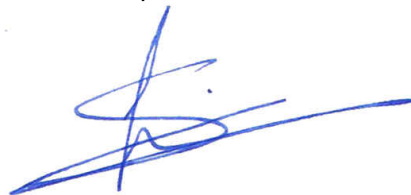
Module 5 - Part 2 : DevOps and Pipeline Security

Date of the course : from 21.04.2026 to 27.04.2026

Hours of training : 14 (100 %)

Attendance rate : 100 %

Esch-sur-Alzette: 29 April 2026



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Module 5 - Part 2 : DevOps and Pipeline Security

Content:

Introduction and Environment Setup

DevSecOps Overview and Key Concepts

- Introduction to DevSecOps and its importance in modern development pipelines
- DevOps vs DevSecOps: The shift-left approach
- Security as Code: Embedding security throughout the CI/CD pipeline
- Compliance, governance, and security controls

CI/CD Pipeline Setup

Hands-on setup of a CI/CD pipeline using Jenkins/GitLab CI/ArgoCD:

- Install and configure Jenkins/GitLab/ArgoCD
 - Build a basic pipeline
 - Integrating version control (Git)
- Secure CI/CD practices:

- Hardening Jenkins and GitLab CI
- Secrets management with Vault/KMS
- Configuring role-based access control (RBAC)

Secure Build and Deployment

Implement static code analysis tools (SonarQube, OWASP):

- Integrating with the pipeline for security analysis
 - Custom rules to detect security misconfigurations
- Container security scanning:

- Using Trivy to scan Docker images for vulnerabilities
- Automating scans as part of the pipeline

Messaging Queue (Kafka/RabbitMQ) Security and Integration

Hands-on setup of Kafka or RabbitMQ:



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- Deploy Kafka/RabbitMQ in a Kubernetes cluster
 - Securing Kafka brokers, Zookeeper, and communications (TLS, ACLs)
- Misconfigurations in message queues:
- Identifying common Kafka/RabbitMQ security risks (e.g., open brokers, no encryption)
 - Using Kafka to send/receive logs for monitoring
- Integrating Kafka with Grafana for monitoring:
- Setting up a dashboard to visualize Kafka metrics
 - Identifying Kafka-related security incidents via Grafana

Advanced Topics and Hands-on Labs

Infrastructure as Code (IaC) Security

Introduction to Terraform/Ansible/Pulumi:

- How IaC fits into the DevSecOps pipeline
 - Writing and deploying secure infrastructure code
- Hands-on demo:

- Deploying infrastructure using Terraform
- Validating security posture with tools like Checkov, TFLint, or Terrascan

CI/CD pipeline integration for IaC security checks

Runtime Security

Security at runtime with Kubernetes:

- Implementing network policies, pod security policies, and RBAC in Kubernetes
- Best practices for securing Kubernetes deployments
- Monitoring runtime security using Falco

Hands-on demo:

- Deploying and securing Kubernetes workloads with Pod Security Standards (PSS)
- Detecting security violations and misconfigurations in real-time with Falco

Incident Detection and Response

Introduction to monitoring and alerting in DevSecOps:

- Setting up monitoring tools (Grafana, Prometheus)



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- Alerting with Alertmanager and PagerDuty
- Detecting and responding to incidents:

- Configuring security alerts for real-time threat detection
- Correlating Kafka logs with security incidents
- Incident response workflow in a DevSecOps pipeline
- Hands-on lab: Set up a secure monitoring and alerting solution

Advanced DevSecOps Best Practices

Hardening and securing microservices:

- Service Mesh (Istio/Linkerd) for security and observability
- Enforcing zero-trust policies across services
- Managing secrets securely (HashiCorp Vault)

Misconfiguration detection using Grafana:

- Setting up real-time dashboards to visualize security misconfigurations
- Automated alerts for detecting configuration drifts

Hands-on lab:

- Simulating security misconfigurations in Kafka/RabbitMQ and Kubernetes
- Detecting and visualizing issues with Grafana dashboards



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Learning Outcomes:

- Build and secure CI/CD pipelines with static code analysis and container scanning.
- Secure message queues (Kafka/RabbitMQ) and monitor them using Grafana.
- Implement infrastructure as code (IaC) security using Terraform and automate checks.
- Harden Kubernetes deployments and detect runtime security issues.
- Set up incident detection, monitoring, and alerting systems for proactive security.



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